

FIITJEE NSEC MOCK TEST-1

Time Allotted: 2 Hours

CODE:

Maximum Marks: 216

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

Write the question paper code (mentioned above) on YOUR OMR Answer Sheet (in the space provided), otherwise your Answer Sheet will NOT be evaluated. Note that the same Question Paper Code appears on each page of the question paper.

Instructions to Candidates:

1. Use of mobile phone, smart watch, and ipad during examination is STRICTLY PROHIBITED.
2. In addition to this question paper, you are given OMR Answer Sheet along with Candidate's copy.
3. On the OMR sheet, make all the entries carefully in the space provided **ONLY** in **BLOCK CAPITALS** as well as by properly darkening the appropriate bubbles.

Incomplete/ incorrect/ carelessly filled information may disqualify your candidature.

4. On the OMR Answer Sheet, use only **BLUE** or **BLACK BALLPOINT PEN** for making entries and filling the bubbles.
5. **Your Enrollment number entered** on the OMR Answer Sheet shall remain your login credentials means login id and password respectively for accessing your performance / result in NSE.
6. Question paper has two parts. In part A1 (Q. No.1 to 48) each question has four alternatives, out of which only one is correct. Choose the correct alternative (s) and fill the appropriate bubble (s), as shown.

Q. No. 22	☐ a	☒	☐ c	☐ d
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In part A2 (Q. No. 49 to 60) each question has four alternatives out of which any number of alternative (s) (1, 2, 3, or 4) may be correct. You have to choose all correct alternative(s) and fill the appropriate bubble(s), as shown

Q. No. 54	☐ a	☒	☐ c	☒
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7. For **Part A1**, each correct answer carries 3 marks whereas 1 mark will be deducted for each wrong answer. In **Part A2**, you get 6 marks if all the correct alternatives are marked. No negative marks in this part.
8. Rough work may be done in the space provided. There are 15 printed pages in this paper
9. Use of **non - programmable scientific** calculator is allowed.
10. No candidate should leave the examination hall before the completion of the examination.
11. After submitting answer paper, take away the question paper & Candidate's copy OMR sheet for your reference.

Please DO NOT make any mark other than filling the appropriate bubbles properly in the space provided on the OMR answer sheet.

OMR answer sheets are evaluated using machine, hence CHANGE OF ENTRY IS NOT ALLOWED. Scratching or overwriting may result in a wrong score.

DO NOT WRITE ON THE BACK SIDE OF THE OMR ANSWER SHEET.

Name of the Candidate : _____

Batch : _____ Date of Examination : _____

Enrolment Number : _____

INDIAN ASSOCIATION OF PHYSICS TEACHERS
NATIONAL STANDARD EXAMINATION IN CHEMISTRY
(NSEC-2023)

Time: 120 minutes

Max. Marks: 216

Attempt All Sixty Questions

A – 1

ONLY ONE OUT OF FOUR OPTIONS IS CORRECT BUBBLE THE CORRECTION OPTION

- We have a solution of NH_4Cl . What effect will addition of NaCl have on the pH of the solution?
(A) Increase pH (B) Decrease pH
(C) No effect (D) Cannot tell from the given information
- A sample of 100 ml of 0.10 M acid HA ($K_a = 1 \times 10^{-5}$) is titrated with standard 0.10 M KOH . How many ml of KOH is required to be added when the pH in the titration flask will be 5.00?
(A) 0 ml (B) 10 ml
(C) 100 ml (D) 50 ml
- K_{sp} of CoS is $4 \times 10^{-12} \text{ M}^2$. Which of the following statements are correct?
(I) Solubility of CoS is $2 \times 10^{-6} \text{ m/L}$.
(II) Solubility of CoS is greater than $2 \times 10^{-6} \text{ m/L}$ because of the hydrolysis of S^{2-} .
(III) Solubility of CoS is greater in an acidic solution than that in pure water.
(IV) Solubility of CoS is decreased when it is simultaneously dissolved along with MnS .
(A) I and III (B) I, III and IV
(C) II, III and IV (D) I and IV
- A weak acid HX has the dissociation constant $1 \times 10^{-5} \text{ M}$. It forms a salt NaX on reaction with alkali. The degree of hydrolysis of 0.1 M solution of NaX is
(A) 0.0001% (B) 0.01%
(C) 0.1% (D) 0.15%
- Following reaction occurs at 25°C ,
 $2\text{NO}(\text{g})(1 \times 10^{-5} \text{ atm}) + \text{Cl}_2(\text{g})(1 \times 10^{-2} \text{ atm}) \rightleftharpoons 2\text{NOCl}(1 \times 10^{-2} \text{ atm})$; the value of ΔG° is
(A) -45.65 kJ (B) -28.53 kJ
(C) -22.82 kJ (D) -57.06 kJ
- Reaction, $\text{A} + \text{B} \rightleftharpoons \text{C} + \text{D}$, is reversible. If for the reaction, $\text{A} + \text{B} \longrightarrow \text{C} + \text{D}$, $\Delta G = -\text{ve}$ then ΔG for $\text{C} + \text{D} \longrightarrow \text{A} + \text{B}$, is
(A) +ve
(B) -ve
(C) +ve and -ve both
(D) -ve for those reactions where K_c for the reaction, $\text{A} + \text{B} \rightleftharpoons \text{C} + \text{D}$, is smaller than 1.

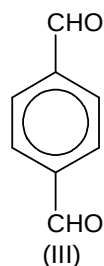
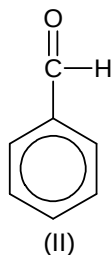
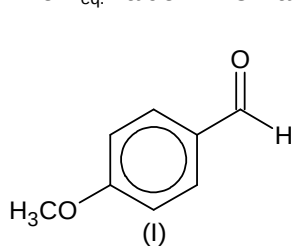
Space for rough work

7. One mole of an anhydrous salt AB dissolves in water with the evolution of 21 Jmol^{-1} of heat. If the heat of hydration of AB is -29.4 Jmol^{-1} , then the heat involved in the conversion of $A^+(g)$ and $B^-(g)$ into AB(s) is
 (A) 50.4 Jmol^{-1} (B) 8.4 Jmol^{-1}
 (C) -50.4 Jmol^{-1} (D) -8.4 Jmol^{-1}
8. The heat of combustion of yellow phosphorus and red phosphorus are -9.91 kJ/mol and -8.78 kJ/mol respectively. The heat of transition from yellow phosphorus to red phosphorus is
 (A) -1.13 kJ (B) -18.69 kJ
 (C) $+18.69 \text{ kJ}$ (D) $+1.13 \text{ kJ}$
9. A 3 mol sample of a triatomic ideal gas at 300 K is allowed to expand under isobaric adiabatic condition from 5 L to 40 L. The value of ΔH is
 (A) -12.46 KJ (B) -14.965 KJ
 (C) -24.62 KJ (D) -10.24 KJ
10. The ratio of van't Hoff factor, if PBr_5 and PCl_5 are assumed to be soluble in water and remains same as in solid state, is
 (A) 2 (B) 1
 (C) 1.5 (D) 2.5
11. $\text{K}_4[\text{Fe}(\text{CN})_6]$ may have the same freezing point at equal concentration of which of the following solution.
 (A) $\text{K}_3\text{Fe}(\text{CN})_6$ (B) MgSO_4
 (C) $\text{Ba}_3(\text{PO}_4)_2$ (D) NaCl
12. For the given electrolyte A_xB_y . The degree of dissociation ' α ' can be given as
 (A) $\alpha = \frac{i-1}{(x+y-1)}$ (B) $i = (1-\alpha) + x\alpha + y\alpha$
 (C) $\alpha = \frac{1-i}{(1-x-y)}$ (D) all of these
13. For the cell reaction,

$$\underset{C_1}{\text{Cu}^{2+}(\text{aq})} + \text{Zn(s)} \rightleftharpoons \underset{C_2}{\text{Zn}^{2+}(\text{aq})} + \text{Cu(s)}$$
 of an electrochemical cell, the change in free energy, ΔG° , at a given temperature is a function of
 (A) $\ln C_1$ (B) $\ln \left(\frac{C_2}{C_1} \right)$
 (C) $\ln C_2$ (D) $\ln (C_1 + C_2)$

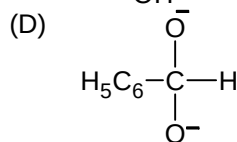
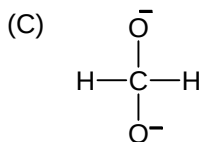
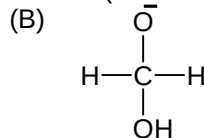
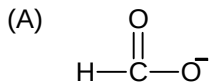
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14. Consider following reaction,
 $\text{Zn} + 2\text{H}^+ \longrightarrow \text{Zn}^{2+} + \text{H}_2$
 half-life period is independent of concentration of Zn at constant pH. At constant Zn concentration, half-life is 10 minutes at pH = 2 and half-life is 100 minutes at pH = 3. Hence, rate law is
 (A) $k [\text{Zn}] [\text{H}^+]$ (B) $k [\text{Zn}] [\text{H}^+]^2$
 (C) $k [\text{Zn}]^0 [\text{H}^+]$ (D) $k [\text{Zn}]^0 [\text{H}^+]^2$
15. What will be the solubility of molar solution of AgCl at 25°C if specific conductivity of saturated AgCl solution is $2.3 \times 10^{-6} \text{ S.cm}^{-1}$ at 25°C and molar conductivities of Ag^+ and Cl^- at infinite dilution are 61.9 and $76.3 \text{ Scm}^2\text{mol}^{-1}$ respectively?
 (A) $2.382 \times 10^{-3} \text{ M}$ (B) $1.66 \times 10^{-5} \text{ M}$
 (C) $5 \times 10^{-8} \text{ M}$ (D) $7.2 \times 10^{-6} \text{ M}$
16. The K_{eq} value in HCN addition to the following aldehydes are in the order:

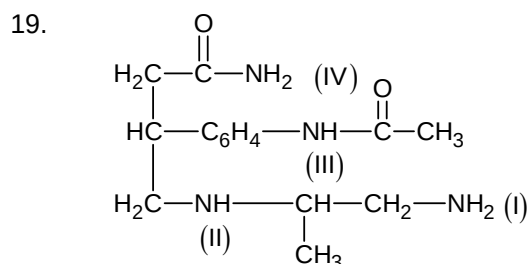
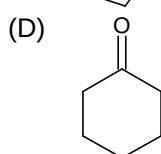
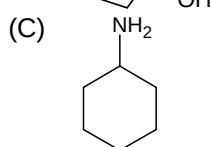
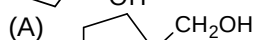
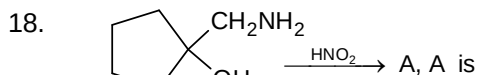


- (A) $\text{I} > \text{II} > \text{III}$ (B) $\text{III} > \text{II} > \text{I}$
 (C) $\text{III} > \text{I} > \text{II}$ (D) $\text{II} > \text{I} > \text{III}$

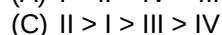
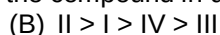
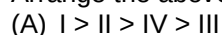
17. In Cannizaro reaction which species is best hydride donor (in conc. Alkaline medium)



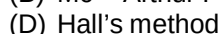
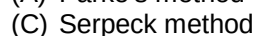
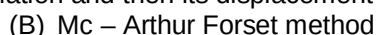
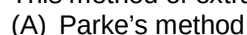
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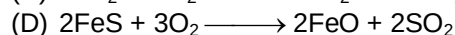
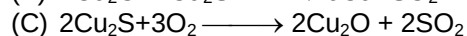
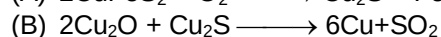
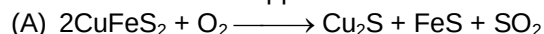
Arrange the above mentioned nitrogen atoms of the compound in decreasing order of basic nature?



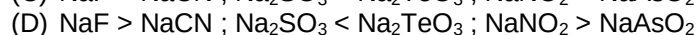
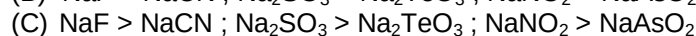
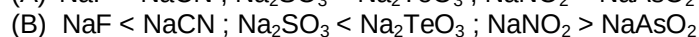
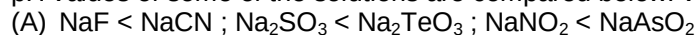
This method of extraction of Ag by complex formation and then its displacement is called:



21. In the extraction of copper the reaction which takes place in Bessemer converter is



22. pH values of some of the solutions are compared below. Which of the following statements is correct?



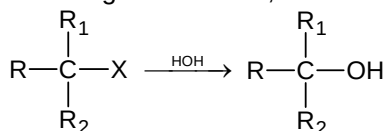
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23. The incorrect statement regarding ClO_2 molecule is that
(A) it contains odd number of electrons and hence paramagnetic.
(B) ClO_2 dimerises.
(C) the Cl – O bond length is shorter than the single Cl – O bond length.
(D) it is a highly reactive species.
24. Which pair is amphoteric?
(i) $\text{Zn}(\text{NH}_2)_2$ (ii) $\text{Al}(\text{OH})_3$
(iii) $\text{Zn}(\text{OH})_2$ (iv) $\text{Be}(\text{OH})_2$
(A) (ii), (iii), (iv) (B) (i), (ii), (iii)
(C) (i), (ii), (iii), (iv) (D) (ii), (iv)
25. The hydroxide which has lowest value of K_{sp} at 25°C is
(A) $\text{Be}(\text{OH})_2$ (B) $\text{Ba}(\text{OH})_2$
(C) $\text{Ca}(\text{OH})_2$ (D) $\text{Mg}(\text{OH})_2$
26. A compound which can be used in space vehicles both to absorb CO_2 and liberate O_2 is
(A) NaOH (B) NaO
(C) Na_2O_2 (D) CaO & NaOH
27. Select the incorrect statement about the compound $\text{NO}[\text{BF}_4]$
(A) it is a diamagnetic species
(B) it has 5σ and 2π bond
(C) B – F bond length in compound is larger than BF_3
(D) nitrogen – oxygen bond is longer than in nitric oxide (NO)
28. Which of the following is tetrahedral and paramagnetic?
(I) MnO_4^- (II) CrO_4^{2-}
(III) $\text{Cr}_2\text{O}_7^{2-}$ (IV) MnO_4^{2-}
(V) ClO_4^- (VI) ClO_3^-
(VII) NO_2 (VIII) N_2O_4
(A) (I), (II) and (III) (B) (IV), (V) and (VI)
(C) (IV) only (D) (V) only
-

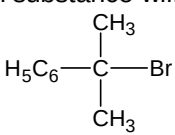
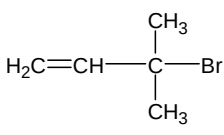
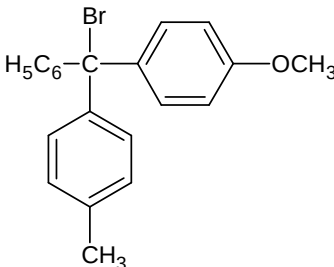
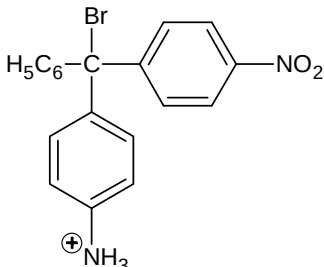
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29. Which of the following order are correct?
- I. Acidity order : o-nitrobenzoic acid > p-nitrobenzoic acid > m-nitrobenzoic acid
- II. Basicity order : $\text{NH}_2^- > \text{EtO}^- > \text{OH}^- > \text{RCOO}^- > \text{Cl}^-$
- III. Heat of hydrogenation : cis-2-butene > trans-2-butene
- IV. Ease of decarboxylation :
- $$\text{C}_6\text{H}_5\text{COCO}_2\text{H} > \text{Ph}-\overset{\text{O}}{\underset{\text{O}}{\text{C}}}-\text{CH}_2-\text{CO}_2\text{H} > \text{Ph}-\underset{\text{OH}}{\text{CH}}-\text{CO}_2\text{H} > \text{Ph}-\underset{\text{NH}_2}{\text{CH}}-\text{CO}_2\text{H}$$
- (A) I and II
(B) I and III
(C) I and IV
(D) I, II and III

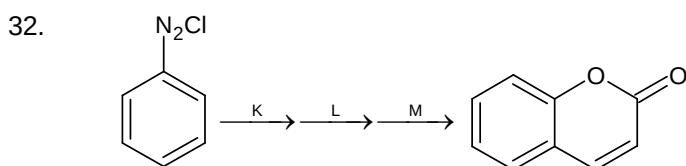
30. For the given reaction,



Which substance will give maximum racemisation?

- (A) 
- (B) 
- (C) 
- (D) 

31. A mixture of gas contains N_2 and C_2H_2 . 20 ml of this mixture is added to 70 ml of O_2 and combustion is allowed to take place over mercury. After cooling the volume of the gas is found to be 68.5 ml. when the resultant gas mixture is passed through KOH solution. The volume of the residual gas is 38.5 ml. The percentage composition of N_2 in the mixture will be:
- (A) 20
(B) 25
(C) 40
(D) 75



The reagents/conditions K, L and M are respectively

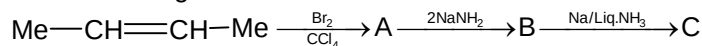
- (A) $\text{Ba}(\text{OH})_2$, KCN and $\text{CHCl}_3/\text{NaOH}$ heat, H_3O^+
- (B) H_2O (boil), CO_2/KOH and $\text{Ac}_2\text{O} - \text{AcONa}$, heat
- (C) Steam ; $\text{CHCl}_3/\text{NaOH}$, heat, H_3O^+ and $\text{Ac}_2\text{O} - \text{AcONa}$, heat
- (D) $\text{Cu}_2\text{Cl}_2/\text{HCl}$, NaOH /high P and CO_2/NaOH , heat

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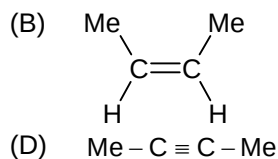
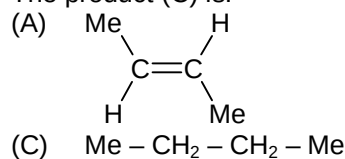
33. A cinema hall has equidistant rows 1m apart. The length of cinema hall is 287m and it has 287 rows. From one side of cinema hall, laughing gas N_2O is released and from the other side, weeping gas ($C_6H_5COCH_2Cl$) is released. In which rows, spectators will be laughing and weeping simultaneously?
 (A) 100th row from weeping gas side (B) 100th row from laughing gas side
 (C) 187th row from the weeping gas side (D) 185th row from the laughing gas side
34. An ionic crystal of AB_2 type crystallizes in fluorite style with density 6.45 g ml^{-1} . Edge length of unit cell is $400 \text{ \AA}$. If all the ions lying on a body diagonal are removed from all the unit cells then what will be the new density? ($B = 19$).
 (A) 6.45 g ml^{-1} (B) 5.3 g ml^{-1}
 (C) 3.3 g ml^{-1} (D) 4.5 g ml^{-1}
35. Which one of the following statements is incorrect?
 (A) Carbon is unaffected by alkalis
 (B) Diamond is unaffected by concentrated acids
 (C) α -Graphite is in the form of ABCABCABC.....
 β -Graphite is in the form of ABABAB.....
 (D) Graphite reacts with F_2 at 500°C , forming intercalation compounds or graphite fluoride $(CF)_n$
36. Argon crystallizes in a structure in which the atoms are located at the following coordinates:
 $(0,0,0); \left(0, \frac{1}{2}, \frac{1}{2}\right); \left(\frac{1}{2}, 0, \frac{1}{2}\right); \left(\frac{1}{2}, \frac{1}{2}, 0\right)$
 The unit cell is:
 (A) simple cubic (B) b. c. c.
 (C) f. c. c. (D) h. c. p
37. Which of the following has different colour than others?
 (A) AgI (B) Ag_2CO_3
 (C) Ag_3PO_4 (D) Ag_2S
38. At 20°C , the osmotic pressure of a solution of 100 ml of which contains 6.33 g of the colouring substance of the blood haematin is 243.4 kPa . What is the molecular formula of the haematin; if the elementary composition of haematin is C = 64.4%, H = 5.2%, N = 3.8%, O = 12.6% and Fe 8.8%?
 (A) $C_{34}H_{33}N_4O_5Fe$ (B) $C_{33}H_{32}N_4O_5Fe$
 (C) $C_{34}H_{30}N_3O_5Fe$ (D) $C_{32}H_{28}N_4O_5Fe$
39. $X (C_9H_{14}) \xrightarrow[CH_2Cl_2]{O_3} 1 \text{ mole ethanal} + 2 \text{ mole glycerol} + 1 \text{ mole Propanal}$
 (one mole)
 No. of stereo isomers possible for compound X are:
 (A) 8 (B) 6
 (C) 4 (D) 10

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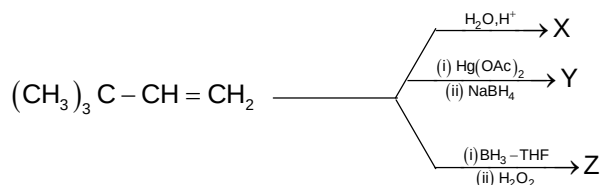
40. In the following reactions:



The product (C) is:



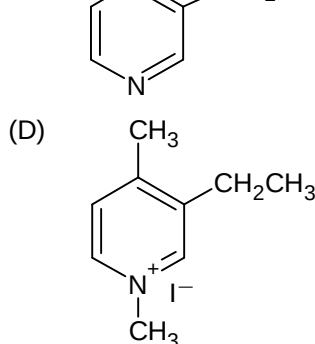
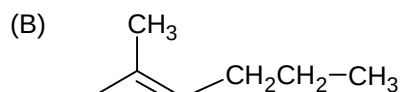
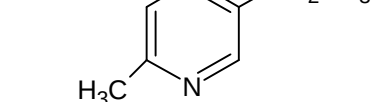
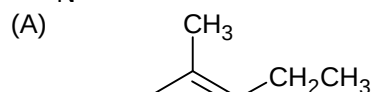
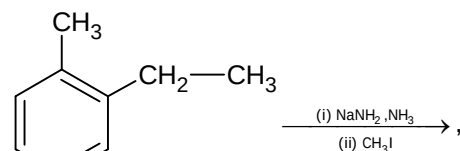
41.



The correct statement from the above sequence is:

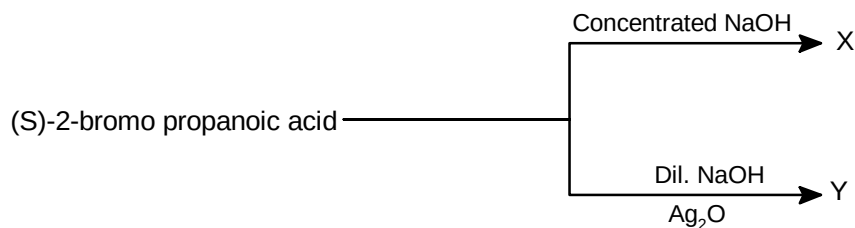
- (A) X and Y are optically active where as Z is inactive
 (B) Y and Z are optically active where as X is inactive
 (C) X and Z are optically inactive where as Y is active
 (D) Y and Z are optically inactive where as X is active

42.



Space for rough work

43.

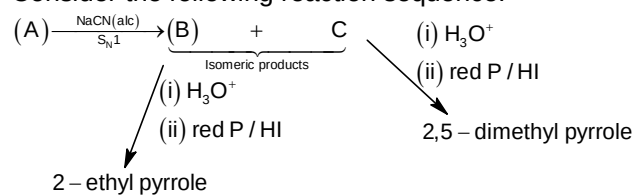


In the above sequence of reactions X and Y are respectively:

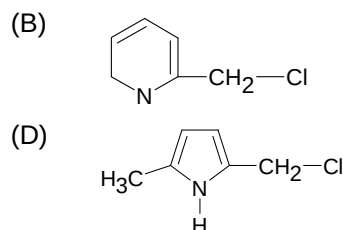
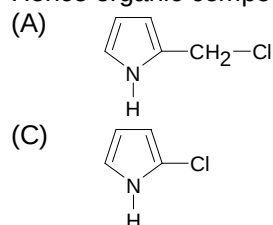
- (A) (S) – Lactic acid and (R) – Lactic acid (B) (R) – Lactic acid and (S) – Lactic acid
 (C) (S) – Lactic acid and (S) – Lactic acid (D) (R) – Lactic acid and (R) – Lactic acid

44.

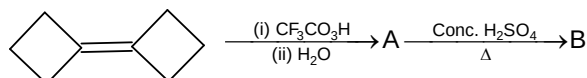
Consider the following reaction sequence:



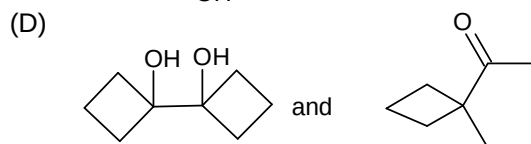
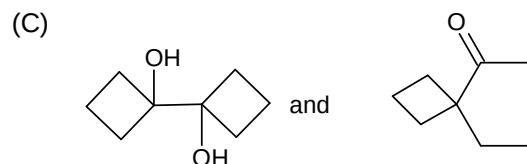
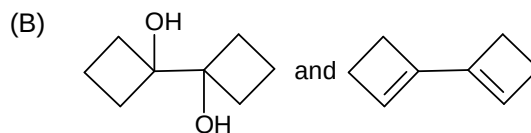
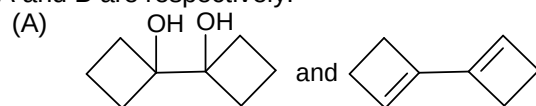
Hence organic compound (A) may be:



45.

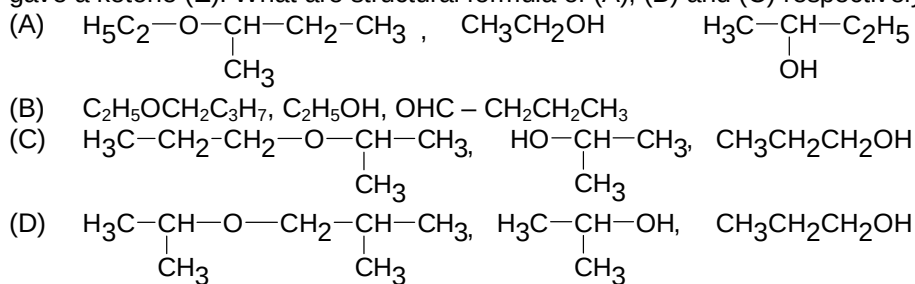


A and B are respectively.

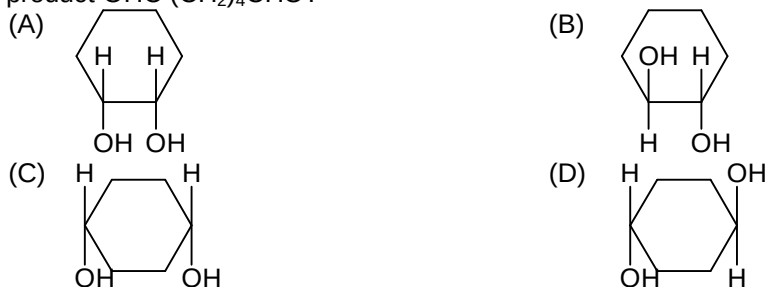


Space for rough work

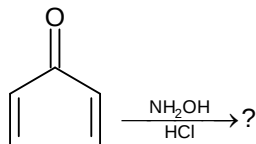
46. An ether (A) $C_6H_{14}O$ when heated with excess of hot conc. HI produced two alkyl halides which on hydrolysis form compounds (B) and (C). Oxidation of (B) gave an acid (D) where as oxidation of (C) gave a ketone (E). What are structural formula of (A), (B) and (C) respectively?



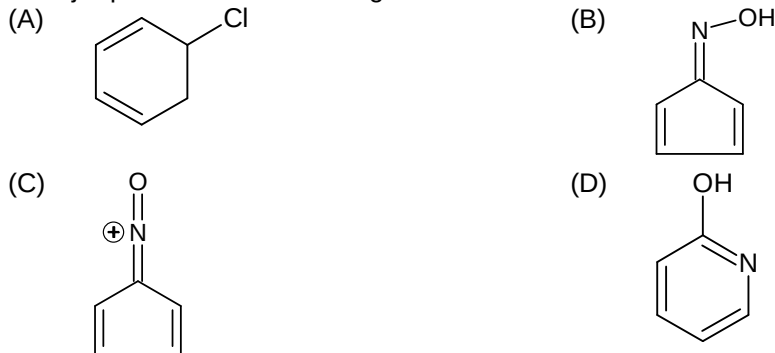
47. Which of the following compounds on oxidation with $(CH_3COO)_4Pb$ would most readily give the product $OHC(CH_2)_4CHO$?



48.

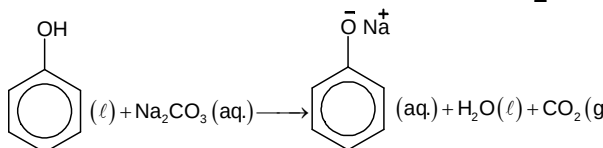


The major product of the following reaction is.....?



Space for rough work

ANY NUMBER OF OPTIONS 4, 3, 2 or 1 MAY BE CORRECT
 MARKS WILL BE AWARDED ONLY IF ALL THE CORRECT OPTIONS ARE BUBBLED

49. Select the correct statement(s) regarding $3P_y$ orbital:
 (A) total number of nodes are 2
 (B) number of maxima in the curve $4\pi r^2 R^2(r)$ Vs r is two
 (C) quantum number n , l and m for orbital may be 3, 1 and -1 respectively
 (D) the magnetic quantum number may have a positive value
50. In which of the following pairs of species the central atom has same hybridization?
 (A) ClF_3O , ClF_3O_2 (B) $(\text{ClF}_2\text{O})^+$, $(\text{ClF}_4\text{O})^-$
 (C) ClF_3 , ClF_3O (D) $(\text{ClF}_4\text{O})^-$, (XeOF_4)
51. Which of the following reaction will not be feasible?
 (A) $\text{R}-\text{C}\equiv\text{CH}(\ell) + \text{Na}(\text{s}) \longrightarrow \text{R}-\text{C}\equiv\text{C}^-\text{Na}^+ + \frac{1}{2}\text{H}_2(\text{g})$
 (B) 
 (C) $\text{R}-\text{C}\equiv\text{CH}(\ell) + \text{NaOH}(\ell) \longrightarrow \text{R}-\text{C}\equiv\text{C}^-\text{Na}^+(\text{aq.}) + \text{H}_2\text{O}(\ell)$
 (D) $\text{R}-\text{C}\equiv\text{CH}(\ell) + \text{NaNH}_2(\text{s}) \longrightarrow \text{R}-\text{C}\equiv\text{C}^-\text{Na}^+(\text{aq.}) + \text{NH}_3(\text{g})$
52. Which of the following statements is/are correct?
 (A) In Serpeck's process for the purification of bauxite, ammonia is obtained as a by product
 (B) In the Hall's process for the purification of bauxite, the ore is fused with NaOH
 (C) In Hall-Heroult's process for the reduction of alumina, a mixture of alumina, cryolite and fluorspar is used
 (D) In the Bayer's process for the purification of bauxite, the ore is digested with Na_2CO_3
53. Electrolysis of molten MgCl_2 is the final production step in the isolation of magnesium from sea water by Dow's process. Assuming that 35.6 g of Mg metal forms. Which of the following statement (s) is/are correct?
 (A) 2.96 moles of electrons are required
 (B) 3.83×10^5 C charge is required
 (C) 31.8 A of current is required for 2.5 hours
 (D) 41.5 A of current is required for approximately 115 mins
54. Which of the following statement(s) is/are correct?
 (A) Pot.tetracyanocuprate(I) is tetrahedral complex
 (B) Sod.tetracyanonickelate(II) is a square planar complex
 (C) Pot.tetrachloridoplatinate(II) is a tetrahedral complex
 (D) Tetraamminecopper(II)sulphate is a square planar complex

Space for rough work

55. Which of the following statement(s) is/are correct?
 (A) At Boyle's temperature a real gas behaves as ideal gas for a wide range of pressure.
 (B) At Boyle's temperature van der Waal's constant 'a' and 'b' compensate each other.
 (C) The area under the Maxwell distribution molecular speed curve remains same irrespective of temperature of gas.
 (D) The fraction of molecules with most probable speed increases with increase of temperature.
56. A saturated solution of AgCl was treated with 1.5 M Na₂CO₃ solution giving Ag₂CO₃. The remaining solution contained 0.0026 gm of Cl⁻ per litre. [K_{sp} of Ag₂CO₃ = $8.2 \times 10^{-12} \text{ M}^3$]. Which of the following statements is/are correct.
 (A) The concentration of NaCl at equilibrium is $7.32 \times 10^{-5} \text{ M}$.
 (B) The concentration of Ag⁺ at equilibrium is $2.33 \times 10^{-6} \text{ M}$.
 (C) K_{sp} of AgCl from the above data is $1.71 \times 10^{-10} \text{ M}^2$
 (D) The equilibrium concentration of CO₃⁻² is $8.2 \times 10^{-12} \text{ M}$
57. Which of the following pairs can be distinguished by Fehling's solution?
 (A) Glucose and fructose
 (B) Glucose and sucrose
 (C) Benzaldehyde and ethanal
 (D) Hydroxy propanone and benzaldehyde
58. Which of the following dicarboxylic acids on heating form anhydride?
 (A) malonic acid
 (B) succinic acid
 (C) glutaric acid
 (D) adipic acid
59. $\text{CH}_3 - \underset{\text{O}}{\underset{\parallel}{\text{C}}} - \text{CH}_2 - \text{Ph} \xrightarrow{\text{NOCl}/\text{H}^+} \text{'X'} \xrightarrow{\text{LiAlH}_4} \text{'Y'} \xrightarrow[0-5^\circ \text{C}]{\text{NaNO}_2 + \text{HCl}} \text{'Z'}$
 (A) Compound 'X' is $\text{CH}_3 - \underset{\text{O}}{\underset{\parallel}{\text{C}}} - \underset{\text{NH}_2}{\text{CH}} - \text{Ph}$
 (B) 'Y' is optically active
 (C) Compound 'Z' is $\text{CH}_3 - \underset{\text{O}}{\underset{\parallel}{\text{C}}} - \text{CH}_2 - \text{Ph}$
 (D) Compound 'Y' is $\text{CH}_3\text{CH}_2\underset{\text{NH}_2}{\text{CH}} - \text{Ph}$
60. Which of the following compounds are formed by the oxidation of
 $\begin{array}{c} \text{CH}_3 \\ | \\ \text{H}_3\text{C} - \text{C} - \text{CH} - \text{CH}_2 \\ | \quad | \quad | \\ \text{OH} \quad \text{OH} \quad \text{OH} \end{array}$
 with HIO₄
 (A) CH₃ - CHO
 (B) CH₃COCH₃
 (C) H - COOH
 (D) HCHO

Space for rough work

ANSWERS

- | | | | |
|------------------|---------------|---------------|---------------|
| 1. (C) | 2. (D) | 3. (B) | 4. (B) |
| 5. (A) | 6. (B) | 7. (D) | 8. (A) |
| 9. (B) | 10. (A) | 11. (C) | 12. (D) |
| 13. (B) | 14. (B) | 15. (B) | 16. (B) |
| 17. (C) | 18. (D) | 19. (B) | 20. (B) |
| 21. (B) | 22. (A) | 23. (B) | 24. (C) |
| 25. (A) | 26. (C) | 27. (D) | 28. (C) |
| 29. (D) | 30. (C) | 31. (B) | 32. (C) |
| 33. (A) | 34. (B) | 35. (C) | 36. (C) |
| 37. (D) | 38. (A) | 39. (A) | 40. (A) |
| 41. (C) | 42. (C) | 43. (B) | 44. (A) |
| 45. (C) | 46. (A) | 47. (A) | 48. (D) |
| 49. (A, B, C, D) | 50. (A, C, D) | 51. B, C | 52. (A, C) |
| 53. (A, C, D) | 54. (A, B, D) | 55. (A, B, C) | 56. (A, B, C) |
| 57. (C, D) | 58. (B, C) | 59. (B, C) | 60. (B, C, D) |

HINTS AND SOLUTIONS

1. (C)

2. (D)

$$\begin{aligned} \text{pH} &= \text{pK}_a + \log \frac{[\text{salt}]}{[\text{acid}]} \\ 5 &= 5 + \log \left(\frac{N_2 V_2}{N_1 V_1 - N_2 V_2} \right) \\ 1 &= \frac{0.1 \times V_2}{0.1 \times 100 - 0.1 \times V_2} \\ 10 - 0.1 V_2 &= 0.1 V_2 \\ V_2 &= 50 \text{ ml} \end{aligned}$$

3. (B)

K_{sp} values are experimentally determined. So, they take into account extra solubility due to hydrolysis.

4. (B)

$$\begin{aligned} \text{NaX} &\longrightarrow \text{Na}^+ + \text{X}^- \\ \text{X}^- + \text{H}_2\text{O} &\rightleftharpoons \text{HX} + \text{OH}^- \\ \begin{array}{ccccc} t = 0 & 1.0 & \text{excess} & 0 & 0 \\ & 1 - \alpha & & \alpha & \alpha \\ t = \text{eq.} & 0.1(1 - \alpha) & & 0.1\alpha & 0.1\alpha \end{array} \\ K_h &= \frac{[\text{HX}][\text{OH}^-]}{[\text{X}^-]} \\ &= \frac{[\text{HX}]}{[\text{H}^+][\text{X}^-]} \times \frac{[\text{H}^+][\text{OH}^-]}{1} \\ &= \frac{K_w}{K_a} = \frac{0.1\alpha \times 0.1\alpha}{0.1(1 - \alpha)} \approx 0.1\alpha^2 \quad [1 - \alpha \approx 1 \text{ as } 1 \gg \alpha \text{ for weak acid}] \\ \therefore \frac{10^{-14}}{10^{-5}} &= 0.1\alpha^2 \\ \therefore \alpha &= 1 \times 10^{-4} = 0.01\% \end{aligned}$$

5. (A)

$$\begin{aligned} K_{eq} &= \frac{P_{\text{NOCl}}^2}{P_{\text{NO}}^2 \times P_{\text{Cl}_2}} = 10^8 \\ \therefore \Delta G &= -RT \ln K_{eq} = -2.303 \times 8.314 \times 298 \times 8 = -45.65 \text{ kJ} \end{aligned}$$

6. (B)

7. (D)

8. (A)

$$\begin{aligned} P_{\text{yellow}} &\longrightarrow P_{\text{red}} \\ H_R &= -9.91 - [-8.78] \\ &= -9.91 + 8.78 = -1.13 \text{ kJ} \end{aligned}$$

9. (B)

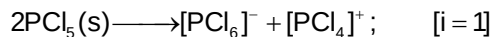
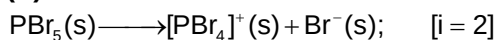
For triatomic gas
 $\gamma = 1.33$, $C_p = 4 R$

$$T_2 = T_1 \left(\frac{V_1}{V_2} \right)^{\gamma-1}$$

$$= 300 \times \left(\frac{5}{40}\right)^{1.33-1} = 300 \times \left(\frac{1}{8}\right)^{1/3} = 300 \times 1/2 = 150$$

$$\Delta H = n C_p (T_2 - T_1) = 3 \times 4 \times 8.314 \times (150 - 300) = -14.965 \text{ KJ}$$

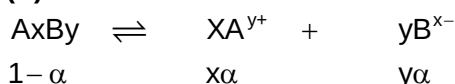
10. (A)



$$\text{ratio} = \frac{2}{1} = 2$$

11. (C)

12. (D)



$$I = (1-\alpha) + x\alpha + y\alpha$$

Hence (D) is correct.

13. (B)

14. (B)

Half-life is independent of concentration of Zn, thus order w.r.t. [Zn] = 1

At pH = 2, $[\text{H}^+] = 10^{-2}$, $t_{1/2} = 10$ minutes

pH = 3, $[\text{H}^+] = 10^{-3}$, $t_{1/2} = 100$ minutes

$$\frac{[t_{1/2}]_1}{[t_{1/2}]_2} = \left(\frac{a_2}{a_1}\right)^{n-1}$$

$$\frac{10}{100} = \left(\frac{10^{-3}}{10^{-2}}\right)^{n-1}$$

$\therefore n = 2$, thus w.r.t. $[\text{H}^+]$ order = 2

Rate law = $k [\text{H}^+]^2 [\text{Zn}]$

Hence, (B) is correct.

15. (B)

$$\begin{aligned} S &= \frac{K \times 1000}{\lambda_{\text{AgCl}}^0} \\ &= \frac{2.3 \times 10^{-6} \times 1000}{(61.9 + 76.3)} \\ &= 1.66 \times 10^{-5} \text{ M} \end{aligned}$$

16. (B)

III > II > I

In I – OCH_3 is electron donating group (+M) so will decrease the electron deficiency at carbonyl carbon to the most.

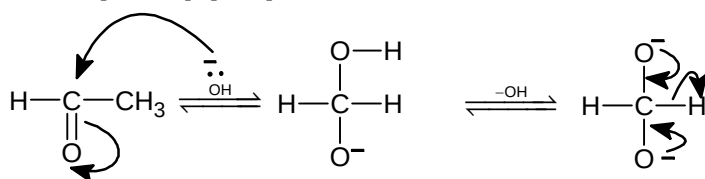
In II there is no electron donating group.

In III electron withdrawing group is present which will increase by electron deficiency at carbonyl carbon.

17. (C)

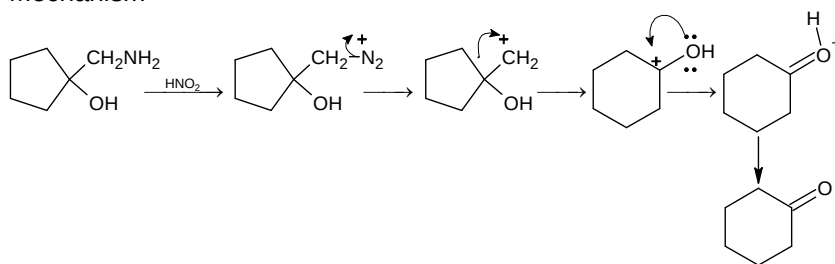
In very conc. base, the rate equation is:

$$\text{rate} = k[\text{HCHO}]^2 [\text{OH}^-]^2$$



The dianion will clearly be a much powerful hydride donor.

18. **(D)**
Mechanism



19. **(B)**

20. **(B)**

21. **(B)**

22. **(A)**
HF is stronger acid than HCN.
H₂SO₃ is stronger acid than H₂TeO₃.
HNO₂ is stronger acid than HAsO₂.

23. **(B)**
It does not dimerise due to delocalization of odd electrons.

24. **(C)**
Zn(NH₂)₂ is amphoteric in NH₃.
Rest are amphoteric in H₂O.

25. **(A)**
Solubility of hydroxides increases down the group, hence Be(OH)₂ is least soluble. Hence, least K_{sp}.

26. **(C)**

27. **(D)**

28. **(C)**

29. **(D)**
- I:

Strong -I effect
-R effect

>

II:

-I effect
-R effect

>

III:

Only -I effect

II: Higher the acidity of an acid lower the basicity of its conjugate base.

III: Higher the stability of an alkene lower the heat of hydrogenation.

IV: Ph-C(=O)-CH₂-COOH is the β-keto acid so it would be having maximum ease for decarboxylation.

30. **(C)**
Since (C) produces the most stable carbocation, hence, this will undergo maximum racemisation.

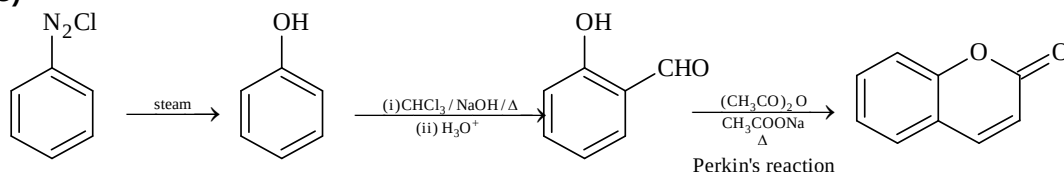
31. **(B)**

$$2\text{C}_2\text{H}_2(\text{g}) + 5\text{O}_2(\text{g}) \longrightarrow 4\text{CO}_2(\text{g}) + \text{H}_2\text{O}(\ell)$$
 Volume of CO₂ = 68.5 – 38.5 = 30 ml
 Volume of C₂H₂ = $\frac{30}{2} = 15\text{ ml}$

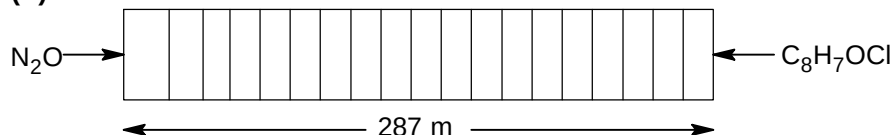
Volume of $N_2 = 20 - 15 = 5 \text{ ml}$

$$\% \text{ of } N_2 = \frac{5}{20} \times 100 = 25\%$$

32. (C)



33. (A)



Molar mass of $C_8H_7OCl = 8 \times 12 + 7 \times 1 + 16 + 35.5 = 154.5 \text{ g}$

Molar mass of $N_2O = 2 \times 14 + 16 = 44 \text{ g}$

According to Graham's Law of diffusion

$$\frac{r_{N_2O}}{r_{C_8H_7OCl}} = \sqrt{\frac{M_{C_8H_7OCl}}{M_{N_2O}}} = \sqrt{\frac{154.5}{44}} = \sqrt{3.5} = 1.87 : 1$$

$$d_{N_2O} = \frac{1.87}{2.87} \times 287 = 187^{\text{th}} \text{ row from } N_2O \text{ side}$$

$$d_{C_8H_7OCl} = \frac{1.0}{2.87} \times 287 = 100^{\text{th}} \text{ row from weeping gas side}$$

Therefore, the spectator from the side of N_2O in the 187^{th} row will be laughing and weeping simultaneously. Alternatively, the spectator from the side of weeping gas in 100^{th} row will laugh and weep.

34. (B)

$$d = \frac{Z \times M}{a^3 \times N_0}$$

$$M = \frac{6.45 \times 64 \times 10^{-24} \times 6.02 \times 10^{23}}{4}$$

($Z = 4$ as it is fluorite type) $M = 62$

Now if all the ions removed from one body diagonal then new mass of effective unit cell = $M' \times Z = 204$

$$d = \frac{Z \times M'}{a^3 \times N_0} = 5.30 \text{ g ml}^{-1}$$

35. (C)

(C) α -Graphite is in the form of ABABAB.....

β -Graphite is in the form of ABCABCABC.....

36. (C)

37. (D)

All others are yellow in colour while Ag_2S is black.

38. (A)

$$\pi = CRT, \text{ So, } C = \frac{\pi}{RT} = \frac{293.4}{8.31 \times 293} = 0.1 \text{ mol/L}$$

If 100 ml contains 6.33 g of haematin.

$\therefore$ 1 L will contain 63.3 g of haematin.

Molecular mass of haematin = 633 u

Estimating the empirical formula of compound

$$C : H : N : O : F = \frac{64.6}{12} : \frac{5.2}{1} : \frac{8.8}{14} : \frac{12.6}{16} : \frac{8.8}{56}$$

$\approx 34 : 33 : 4 : 5 : 1$

So, empirical formula is $C_{34}H_{33}N_4O_5Fe$, whose mass corresponds to the molecular mass.

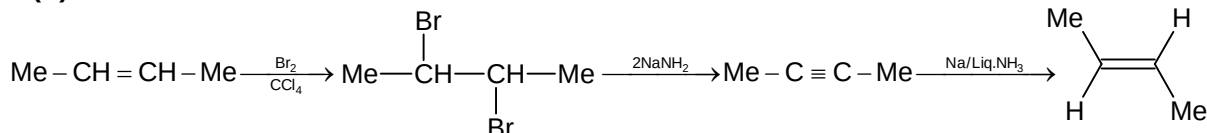
39. (A)



No. of stereo site = 3

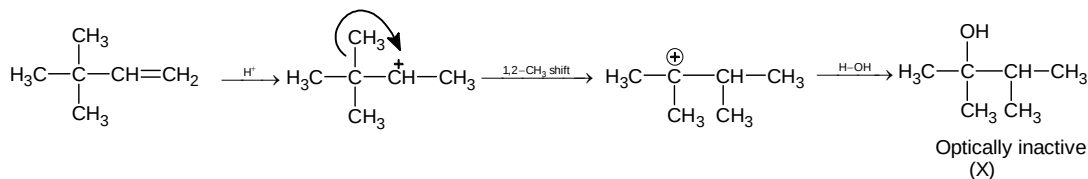
No. of stereo isomer = $2^3 = 8$

40. (A)

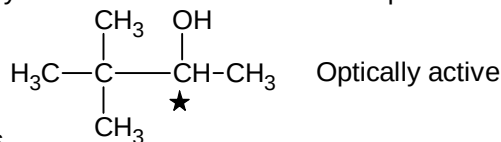


41. (C)

(A)

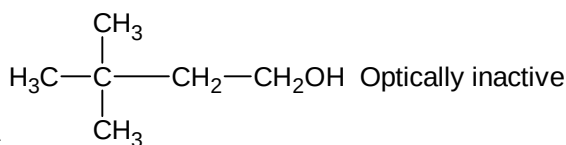


(B) On oxymercuration demercuration the product is alcohol without rearrangement



$\therefore$ Y is

(C) On hydroboration – oxidation the addition is against to Markonikov's rule



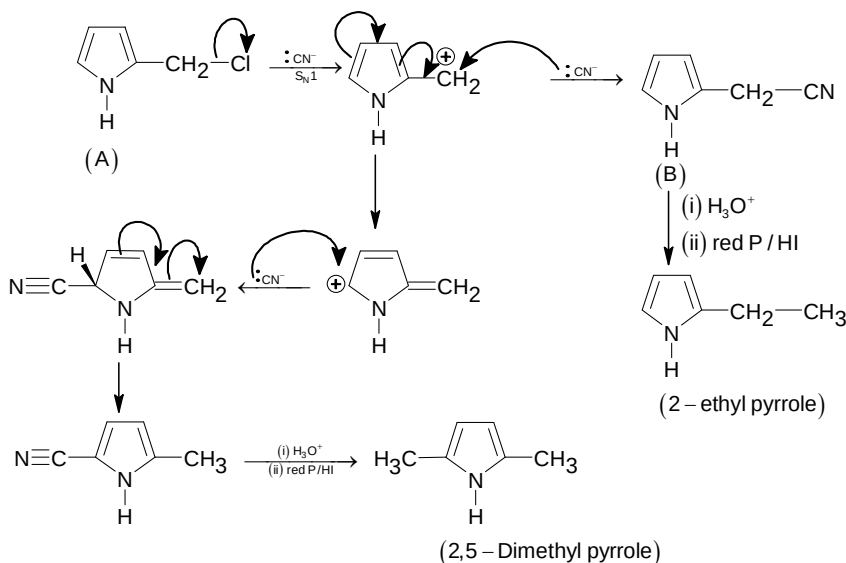
$\therefore$ Z is

42. (C)

43. (B)

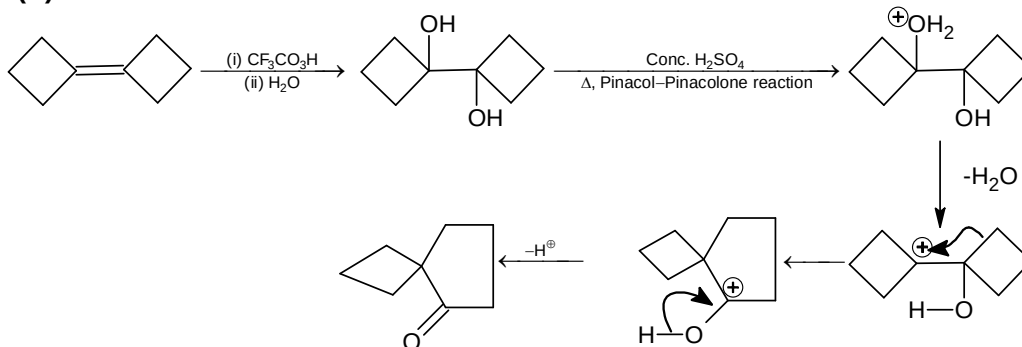
With conc. NaOH the reaction is S_N2 , hence inversion in configuration. With Ag_2O , NaOH, neighbouring group participation. Retention in configuration.

44. (A)

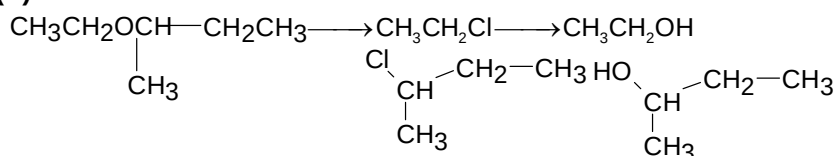


Hence, (A) is the correct answer.

45. (C)

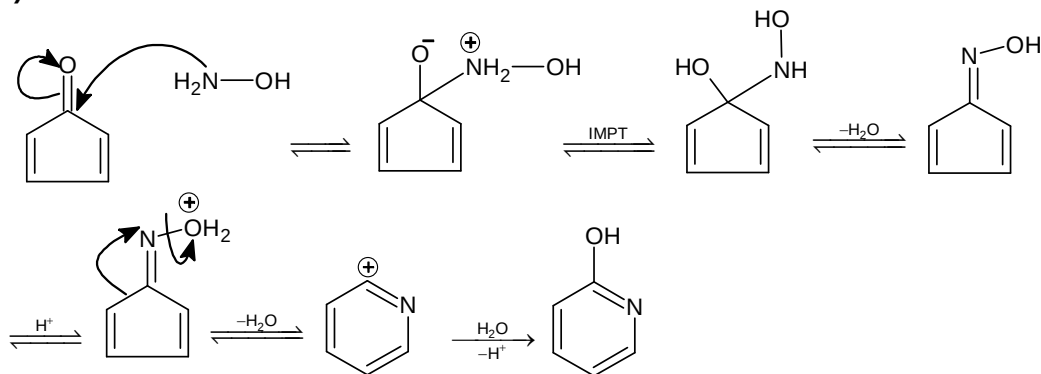


46. (A)



47. (A). A is cis 1, 2-diol.

48. (D)



49. (A, B, C, D)

For $3P_y$ orbital

Total nodes = $n - 1 = 2$

50. (A, C, D)

In all ClF_3O , ClF_3O_2 , ClF_3 hybridisation of Cl is sp^3d and ClF_4O^- and XeOF_4 hybridisation of central atom is sp^3d^2 .

51. (B, C). Weak acid-base pair cannot react to form strong acid-base pair.

52. (A, C)

In the Hall's process for the purification of bauxite the ore is fused with Na_2CO_3 .

In the Bayer's process for the purification of bauxite the ore is digested with NaOH .

53. (A, C, D)

$$\frac{35.6}{12} = 2.966 \text{ equivalents}$$

For 1 equivalent 96500 C is required

So, for 2.966 equivalent 286283.33 C is needed so.

2.97 moles of electron carries same charge hence (A) is correct.

$$Q = It$$

$$Q = 31.8 \times 2.5 \times 60 \times 60$$

$$Q = 286283.33$$

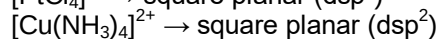
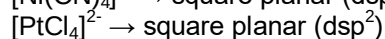
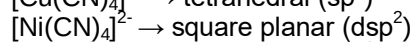
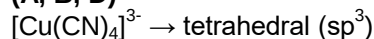
Hence, (C) is correct.

$$Q = 41.5 \times 115 \times 60$$

$$= 286283.33$$

Hence, (D) is correct.

54. **(A, B, D)**



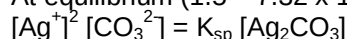
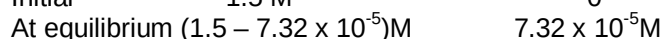
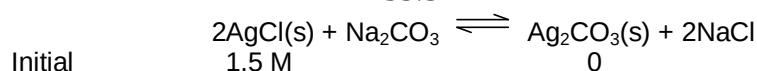
Hence, (A), (B) and (D) are the correct answer.

55. **(A, B, C)**

56. **(A, B, C)**



$$\text{At equilibrium } [\text{Cl}^-] = [\text{NaCl}] = \frac{0.0026}{35.5} = 7.32 \times 10^{-5} \text{ M}$$



$$\therefore [\text{Ag}^+] = \sqrt{\frac{8.2 \times 10^{-12}}{1.5}} = 2.33 \times 10^{-6} \text{ M}$$

$$\therefore K_{\text{sp}} \text{ of AgCl} = [\text{Ag}^+] [\text{Cl}^-] = (2.338 \times 10^{-6}) \times (7.32 \times 10^{-5}) = 1.71 \times 10^{-10}.$$

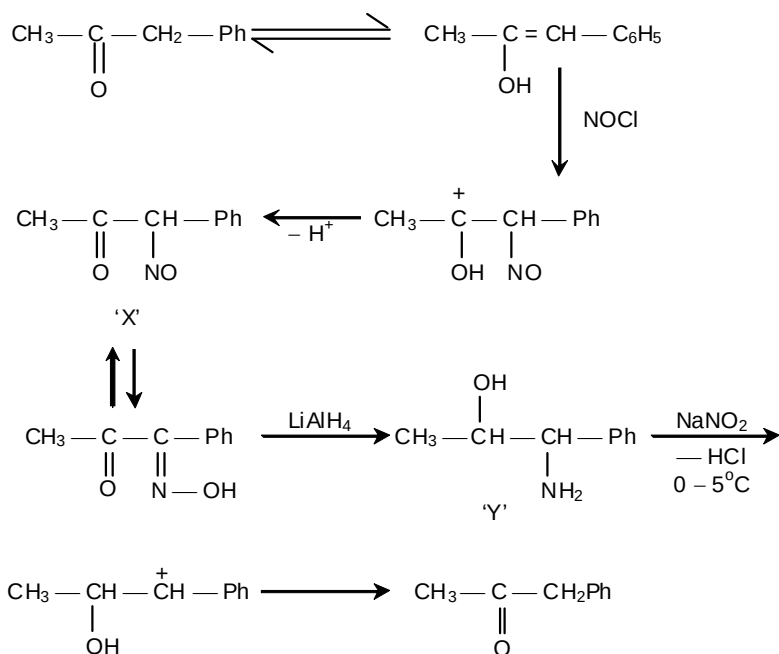
57. **(C, D)**

Glucose, fructose, ethanal and sucrose gives Fehling's test. But HCHO and C₆H₅CHO undergo Cannizzaro reaction in alkaline medium and does not gives Fehling's test.

58. **(B, C)**

According to Blanc's rule only 1,4 & 1,5 dicarboxylic acids can give anhydride

59. **(B, C)**



60. **(B, C, D)**

